Arista Network Access Role QoS Congestion Event SOP (IND-ACC-16)

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8072

Description

This Standard Operating Procedure (SOP) addresses a QoS congestion event on Arista network equipment, specifically for the Access role in the Indiranagar Exchange (IND-ACC-16). The failure domain is characterized by microbursts, interface tail drops, and QoS queue starvation. This issue arises due to the inherent nature of queueing theory and congestion avoidance mechanisms implemented in modern networking protocols such as RSVP-TE and DiffServ. In Arista's EOS, the access role employs a combination of scheduling algorithms (e.g., PQ, LLQ) and policing mechanisms to manage traffic. However, when a sudden surge in traffic occurs (microbursts), the system may experience queue congestion, leading to packet drops and associated QoS issues. Furthermore, inadequate policing or scheduling may exacerbate this problem. This SOP is designed to identify and mitigate such issues by utilizing diagnostic commands and resolution steps tailored to Arista's EOS and the access role.

Pre-requisite Checks

- Verify the Indiranagar Exchange site location and the affected node (IND-ACC-16) is accessible for remote hands.

- Confirm the redundant link status and network topology to ensure no other failures are impacting the network.

- Gather relevant logs and system information from the node using the 'show system internal log' and 'show system internal info' commands.

Potential Root Causes

Cause	Probability
Inadequate policing or scheduling in the access role, leading to QoS queue starvation	High
RSVP-TE or DiffServ configuration issues	Medium
Hardware or software defects in the Arista node (IND-ACC-16)	Low

Diagnostic Commands & Expected Outputs

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Command: show qos queue counters, Expected Output: Queue counters for each interface, including drops and enqueue events, Description: This command provides insight into the QoS queue performance, allowing us to identify potential congestion points.

Command: show interface counters, Expected Output: Interface counters for each interface, including errors and drops, Description: This
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command helps us identify potential issues with the interface, such as hardware defects or misconfiguration.

Command: show system internal log, Expected Output: System log information, including errors and warnings, Description: This command provides detailed system information, including any errors or warnings related to the QoS congestion event.
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Resolution Steps

Step 1: Step: Verify and adjust policing and scheduling configurations, Command: set qos policy policing , Description: Adjust policing and scheduling configurations to ensure adequate QoS queue management and prevent congestion.

Step 2: Step: Review and correct RSVP-TE or DiffServ configurations, Command: show mpls rsvp-te config, Description: Review and correct RSVP-TE or DiffServ configurations to ensure proper traffic management and prevent congestion.

Step 3: Step: Perform hardware and software checks on the Arista node, Command: show system internal info, Description: Perform hardware and software checks on the Arista node to identify and address any defects or misconfigurations.

Post-Check Verification

```
Command: show qos queue counters, Expected Output: Improved queue counters, including reduced drops and enqueue events, Description: Verify that the QoS queue performance has improved after the resolution steps.

Command: show interface counters, Expected Output: Improved interface counters, including reduced errors and drops, Description: Verify that the interface counters have improved after the resolution steps.
```

Roll-back Procedures

- Step: Restore original policing and scheduling configurations, Command: set qos policy policing , Description: Restore the original policing and scheduling configurations if the fix fails.

- Step: Restore original RSVP-TE or DiffServ configurations, Command: show mpls rsvp-te config , Description: Restore the original RSVP-TE or DiffServ configurations if the fix fails.